Summary of agent reports on *Matrix Theory Reloaded: A BPS Road to Holography*

Setup

There are 20 runs in `runs/`: five models (Claude Haiku 5.5, Sonnet 5.5 and Opus 5.5; Gemini 3.8 Flash; Gemini 4 Argon), two effort levels (low, high) and two repeats each. Every run produced a `report.tex` covering (1) the principal achievement, (2) the principal drawback, and (3) the follow-up idea that best balances feasibility and impact. Below, each report's choices are sorted into categories. Where a report lists several items, it is classified by the item it presents as the main one.

1. Principal achievement: unanimous

All **20/20** reports name the same achievement: *conceptual unification*. The matrix-theory (MpT) BPS decoupling limit, applied at asymptotic infinity of a Dp-brane solution, *is* the Maldacena/IMSY near-horizon limit. One framework then links matrix theory, AdS/CFT, non-Lorentzian holography and $T\bar{T}$. Every report also highlights the p-brane $T\bar{T}$ flow equations (free scalars $\to$ Nambu–Goto for all p; Maxwell $\to$ DBI for $p = 1, 2$; the simple 3D M2/D2 flow) as the main *concrete* result. Most also cite the DLCQ^n/DLCQ^m classification as part of the achievement. Only two reports say they checked the formulas themselves: Haiku high 1 (by hand, $p = 1, 2$) and Opus high 1 (numerically, $p = 1, \ldots, 5$).

2. Principal drawback

Almost every report lists much the same set of weaknesses. They differ in which one they put first. The four headline framings are:

K Kinematic, untested holography (8): the DLCQ^n/DLCQ^m conjecture is a classification of limits. No observable (spectrum, correlator, index, thermodynamics) is matched on both sides.

G Unverified non-Lorentzian bulk geometries (7): they are not shown to solve MpT supergravity (equations of motion and torsional constraints). They rely on an ad hoc ℓ/N rescaling or on smearing (Gemini Argon stresses that these two procedures give inequivalent geometries).

C Classical, abelian, probe-level character (4): the flows and bulk constructions lack quantum and non-abelian control, and higher-dimensional $T\bar{T}$ is not well defined quantum mechanically.

T Bulk-from-$T\bar{T}$ is an ansatz (1): the position-dependent coupling $\lambda(X) = \Omega^{-2}$ is assumed, not derived from a flow.

The framing depends on the model family far more than on effort. Opus always uses K, Flash always C, and Argon always G. Table 2 shows that the individual criticisms overlap heavily whatever the headline.

Table 1: Headline drawback by model and effort (counts; 2 runs per cell).

	Low effort				High effort			
Model	K	G	C	T	K	G	C	T
Claude Haiku 5.5	–	2	–	–	1	–	–	1
Claude Sonnet 5.5	2	–	–	–	1	1	–	–
Claude Opus 5.5	2	–	–	–	2	–	–	–
Gemini 3.8 Flash	–	–	2	–	–	–	2	–
Gemini 4 Argon	–	2	–	–	–	2	–	–
Total	4	4	2	0	4	3	2	1

Table 2: Specific weaknesses raised anywhere in each report's drawback section. E: EOM/torsion of non-Lorentzian backgrounds unchecked; R: ad hoc ℓ/N rescaling or smearing; O: no observable matched across a proposed duality; A: flows classical/bosonic/abelian; P: $p \geq 3$ gauge-field flows incomplete; B: Poincaré patch or asymptotic breakdown (strong coupling, α'); H: bulk-from-$T\bar{T}$ heuristic.

Model	Effort	Run	E	R	O	A	P	B	H
Haiku	low	1	✓	✓	✓	✓	✓	✓	
Haiku	low	2	✓	✓	✓	✓		✓	
Haiku	high	1	✓		✓	✓	✓		✓
Haiku	high	2	✓	✓	✓	✓	✓		
Sonnet	low	1	✓	✓	✓	✓		✓	
Sonnet	low	2	✓	✓	✓	✓	✓	✓	
Sonnet	high	1	✓	✓	✓	✓	✓	✓	
Sonnet	high	2	✓	✓	✓	✓	✓	✓	
Opus	low	1	✓	✓	✓	✓		✓	
Opus	low	2	✓	✓	✓	✓		✓	
Opus	high	1	✓	✓	✓	✓	✓	✓	✓
Opus	high	2	✓	✓	✓	✓	✓	✓	
Flash	low	1	✓	✓		✓		✓	
Flash	low	2	✓	✓		✓		✓	
Flash	high	1	✓	✓	✓	✓		✓	
Flash	high	2	✓	✓		✓	✓	✓	
Argon	low	1	✓	✓		✓	✓	✓	✓
Argon	low	2	✓	✓		✓			
Argon	high	1	✓	✓	✓	✓	✓		✓
Argon	high	2	✓	✓		✓	✓		✓
Total (of 20)			20	19	13	20	12	14	5

3. Recommended follow-up direction

The recommendations split evenly. Ten are on the *gravity side*, testing the holographic proposals (S, Q, D). Ten are on the *worldvolume side*, developing the $T\bar{T}$ flows (M, N, P, H).

S Supergravity check of the non-Lorentzian backgrounds (7): formulate (or reduce from 11D non-relativistic supergravity) the MpT field equations, test the proposed solutions, settle smearing vs. N-rescaling, then compute isometries and supersymmetry.

Q Quantitative test of one DLCQ^1/DLCQ^2 pair (2): the M0T limit of the D4 throat vs. DLCQ (2,0) theory/instanton quantum mechanics, by matching symmetry algebras, then a protected spectrum or index.

D D1–D5 testbed (1): a DLCQ^0/DLCQ^2 partial-decoupling limit, with $T\bar{T}$ on the symmetric orbifold CFT and spectrum matching.

M The 3D (p=2) M2/D2 flow (4): study it as a genuine field-theory deformation (quantum solvability, Burgers-type spectrum), make it supersymmetric and non-abelian, and apply it to BLG/ABJM.

N Non-abelian extension of the flows (4), compared with the non-abelian DBI/Myers action: at $p = 1$ or $p = 1, 2$ (Haiku, Sonnet) or for $p = 0$ BFSS/BMN matrix quantum mechanics (Flash).

P Position-dependent flow (1): derive the flow for $\lambda(X)$ and check that it generates the near-horizon background.

H Cutoff holography (1): match the $p = 2$ flow to the radial Hamilton–Jacobi equation of AdS_4 at finite cutoff.

Table 3: Recommended follow-up by model and effort (counts; 2 runs per cell).

	Low effort							High effort						
Model	S	Q	D	M	N	P	H	S	Q	D	M	N	P	H
Haiku	1	–	–	–	1	–	–	1	–	–	–	–	1	–
Sonnet	2	–	–	–	–	–	–	1	–	–	–	1	–	–
Opus	–	–	–	2	–	–	–	–	2	–	–	–	–	–
Flash	–	–	–	–	2	–	–	–	–	1	–	–	–	1
Argon	–	–	–	2	–	–	–	2	–	–	–	–	–	–
Total	3	0	0	4	3	0	0	4	2	1	0	1	1	1

Effort effect. At high effort, 7 of 10 runs recommend a gravity-side test (S/Q/D). At low effort, 7 of 10 recommend developing the flows (M/N). The shift is clearest for Opus (low: p=2 flow; high: M0T/D4 duality test) and Argon (low: non-abelian p=2 flow on ABJM; high: supergravity verification). Both repeats agree in 8 of the 10 model–effort cells. The two exceptions are Haiku high (P vs. S) and Flash high (H vs. D). The supergravity check (S) is the single most common choice (7/20). It also appears as the runner-up or "more ambitious" alternative in most of the remaining reports.

Per-run classification

Style note. The Claude reports are short (4–9 kB of source) and similar in length across efforts. The Gemini reports are longer, especially at high effort (16–21 kB), and use more emphatic language ("landmark", "first rigorous proof", "None" listed as the main bottleneck).

Model	Effort	Run	Drawback	Next	Follow-up in brief
Haiku	low	1	G	N	Non-abelian $p = 1$ flow vs. non-abelian DBI
Haiku	low	2	G	S	EOM check of M2/D2 near-horizon backgrounds
Haiku	high	1	T	P	Position-dependent $p = 1, 2$ flow generating the bulk
Haiku	high	2	K	S	$1/\omega$ expansion of type II equations on one background
Sonnet	low	1	K	S	Formulate bosonic MpT supergravity; verify solutions
Sonnet	low	2	K	S	MpT supergravity test, then isometries and a probe observable
Sonnet	high	1	G	S	MpT supergravity for $p = 0, 1$; test double-limit solutions
Sonnet	high	2	K	N	Non-abelian $p = 1, 2$ flows vs. known F^4, F^6 terms
Opus	low	1	K	M	3D flow: well-posedness, quantum solvability, BLG/ABJM
Opus	low	2	K	M	3D flow: Burgers-type solvability, susy, non-abelian
Opus	high	1	K	Q	M0T/D4 vs. instanton QM: sugra, symmetry, index
Opus	high	2	K	Q	M0T/D4 via null-reduced $\mathrm{AdS}_7 \times S^4$: algebra, KK spectrum
Flash	low	1	C	N	Non-abelian $p = 0$ flow of BFSS matrix QM
Flash	low	2	C	N	Non-abelian $p = 0$ flow for BFSS/BMN; $\alpha'^3 R^4$ matching
Flash	high	1	C	H	$p = 2$ flow vs. AdS_4 cutoff Hamilton–Jacobi evolution
Flash	high	2	C	D	D1–D5 partial decoupling with symmetric-orbifold $T\bar{T}$
Argon	low	1	G	M	Non-abelian, susy $p = 2$ flow applied to 3D SYM and ABJM/BLG
Argon	low	2	G	M	Same, with comparison to M2 amplitudes and AdS_4
Argon	high	1	G	S	Sourced MpT sugra from 11D; localisation puzzle; M0T/D4 probe
Argon	high	2	G	S	11D non-rel. sugra and Killing spinors; smeared vs. rescaled; $B_{ij} \neq 0$